The Independent Set Polytope

$M = (E, \mathcal{I})$

$P_{\text{Ind set}}(M) = P = \text{conv}\{ \chi^I : I \in \mathcal{I} \}$

$Q_{\text{Ind Set}}(M) = Q = \{ x \geq 0,\ x(U) \leq r_M(U),\ \forall U \subseteq E \}$

Thm: $Q = P$

Note: $P \subseteq Q$ immediate.

The Reverse Inclusion

We prove the reverse inclusion by showing that the system is TDI.

For $w \in \mathbb{Z}^n$, $(n = |E|)$

$$\begin{array}{l|l} \max: w^T x & \min: \sum y_U \cdot r_M(U) \\ \text{st: } \sum_{i \in U} x_i \le r_M(U), \ \forall U \subseteq E & \text{st: } \sum_{U \subseteq E :\, i \in U} y_U \ge w_i, \ \forall i \in E \\ \quad x \ge 0 & \quad y_U \ge 0 \quad \forall U \end{array}$$

Ex: We can take $w \in \mathbb{Z}^n_+$ wlog.

Proof — 1/2

Order the elements of E so that

$$w_1 \geq w_2 \geq w_3 \geq \dots \geq w_n \geq 0$$

$U_i = \{1, 2, \dots, i\}$ in this ordering

Let $I = \{ i : r(U_i) = r(U_{i-1}) + 1 \}$

Note: I is the output of the Greedy algorithm

hence I is a max wt ind. set and

hence $x = \chi^I$ is a primal optimal solution.

Proof — 2/2

Let's define a dual solution:

$$w(I) = \sum_{i \in I} w_i = \sum_{i=1}^{n} w_i \left(r(U_i) - r(U_{i-1}) \right)$$

$$= w_n \cdot r(U_n) + \sum (w_i - w_{i+1}) \, r(U_i)$$

$$= \sum_{i=1}^{n} \lambda_i \cdot r(U_i), \qquad \lambda_i = w_i - w_{i+1}, \quad \lambda_n = w_n$$

Let: $y_{U_i} = \lambda_i$, $y_U = 0$

Note: y is dual feasible, $\sum y_U r_M(U) = w(I)$, thus opt.

Hence:

1. Primal & Dual have integral opt solutions (TDI)
2. Dual has an opt. sol'n whose support is laminar

The Base Polytope

Corollary: For $P_{Base}(M) = \text{conv}\{\chi^B : B \text{ is a base}\}$

$$Q_{Base}(M) = \{x \geq 0,\ x(U) \leq r_M(U),\ \forall U \subseteq E,\ x(E) = r_M(E)\}$$

$$P_{Base} = Q_{Base}$$

pf: $P_{Base} \subseteq Q_{Base} \subseteq P_{Ind\,set} = Q_{Ind\,set}$

Take $x \in Q_{Base} \Rightarrow x \in P_{Ind\,set} \Rightarrow x = \sum_{I \in \mathcal{I}} \alpha_I \chi^I$, $\sum \alpha_I = 1$, $\alpha_I \geq 0$.

$$x(E) = \sum_{i=1}^{n} x_i = \sum_{I \in \mathcal{I}} \sum_{i \in I} \alpha_I = \sum_{I \in \mathcal{I}} \alpha_I \cdot \underbrace{|I|} = \sum_{I \in \mathcal{I}} \alpha_I \cdot r(I) \leq \underbrace{\sum_I \alpha_I \cdot r(E)}_{= r(E)}$$

Since $x(E) = r(E) \Rightarrow r(I) = r(E)\ \forall I : \alpha_I > 0$

$\Rightarrow$ Each I s.t. $\alpha_I > 0$ is a base. $\blacksquare$